

Tabla 1 — Esquema de Subredes

Subred	Dirección de Red	Máscara	Rango de Hosts Válidos	Broadcast	Gateway	Uso
Servidores	172.16.0.0	255.255.224.0 (/19)	172.16.0.1 – 172.16.31.254	172.16.31.255	172.16.0.1	Servidores
Suc1				5		SW1
RRHH	172.16.32.0	255.255.224.0 (/19)	172.16.32.1 – 172.16.63.254	172.16.63.255	172.16.32.1	PCs RRHH
Suc1 (Red A)				5		SW2 Fa0/1–0/10
Médicos	172.16.64.0	255.255.224.0 (/19)	172.16.64.1 – 172.16.95.254	172.16.95.255	172.16.64.1	PCs
Suc1 (Red B)				5		Médicos SW2 Fa0/11–0/20
WiFi Suc1 (Red C)	172.16.96.0	255.255.224.0 (/19)	172.16.96.1 – 172.16.127.254	172.16.127.255	172.16.96.1	APs SW2 Fa0/21–0/24 DHCP
Servidores	172.16.128.0	255.255.224.0 (/19)	172.16.128.1 – 172.16.159.254	172.16.159.255	172.16.128.1	Servidores
Suc2				1		SW3
RRHH	172.16.160.0	255.255.224.0 (/19)	172.16.160.1 – 172.16.191.254	172.16.191.255	172.16.160.1	PCs RRHH
Suc2 (Red A)				1		SW4 Fa0/1–0/10

Médicos	172.16.192.0	255.255.224.0 (/19)	172.16.192.1 – 172.16.223.254	172.16.223.255	172.16.192.1	PCs
Suc2 (Red B)	0	0 (/19)	172.16.223.254	55	1	Médicos SW4 Fa0/11–0/20
WiFi Suc2 (Red C)	172.16.224.0	255.255.224.0 (/19)	172.16.224.1 – 172.16.255.254	172.16.255.255	172.16.224.1	APs SW4 Fa0/21–0/24 DHCP

Tabla 2 — Configuración de Servidores

Servidor	IP Asignada	Sucursal	Switch / Piso	Rol / Función	Acceso desde
Srv-SMTP-Suc1	172.16.0.2	1	SW1 – Planta Baja	Envío y recepción de correo interno (clinica.org)	Todas las redes Suc1
Srv-FTP-Suc1	172.16.0.3	1	SW1 – Planta Baja	Transferencia de archivos por rol	Todas las redes Suc1
Srv-DNS-Suc1	172.16.0.4	1	SW1 – Planta Baja	Resolución de nombres internos	Todas las redes
Srv-Web-Clinica	172.16.0.5	1	SW1 – Planta Baja	Página institucional (web-clinica.org)	Todas las redes
Srv-Web-Médicos	172.16.0.6	1	SW1 – Planta Baja	Panel médico (medicos.clinica.org)	Todas las redes

Srv-DHCP-Suc1	172.16.0.7	1	SW1 – Planta Baja	Asignación dinámica de IPs para WiFi Suc1	Red C WiFi Suc1
Srv-Web-RRHH	172.16.0.8	1	SW1 – Planta Baja	Panel RRHH (rrhh.clinica.org)	Todas las redes
Srv-SMTP-Suc2	172.16.128.2	2	SW3 – Planta Baja	Envío y recepción de correo interno (clinica.org)	Todas las redes Suc2
Srv-FTP-Suc2	172.16.128.3	2	SW3 – Planta Baja	Transferencia de archivos por rol	Todas las redes Suc2
Srv-DNS-Suc2	172.16.128.4	2	SW3 – Planta Baja	Resolución de nombres internos Suc2	Todas las redes
Srv-Web-Clinica-S2	172.16.128.5	2	SW3 – Planta Baja	Página institucional (web-clinica.org)	Todas las redes
Srv-Web-Medicos-S2	172.16.128.6	2	SW3 – Planta Baja	Panel médico (medicos.clinica.org)	Todas las redes
Srv-DHCP-Suc2	172.16.128.7	2	SW3 – Planta Baja	Asignación dinámica de IPs para WiFi Suc2	Red C WiFi Suc2
Srv-Web-RRHH-S2	172.16.128.8	2	SW3 – Planta Baja	Panel RRHH (rrhh.clinica.org)	Todas las redes

Tabla 3 — IPs asignadas a PCs

Dispositivo	IP	Máscara	Gateway	DNS	Red
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PC1-RRHH-Suc1	172.16.32.2	255.255.224.0	172.16.32.1	172.16.0.4	Red A – RRHH Suc1
PC2-RRHH-Suc1	172.16.32.3	255.255.224.0	172.16.32.1	172.16.0.4	Red A – RRHH Suc1
PC3-Medico-Suc1	172.16.64.2	255.255.224.0	172.16.64.1	172.16.0.4	Red B – Médicos Suc1
PC4-Medico-Suc1	172.16.64.3	255.255.224.0	172.16.64.1	172.16.0.4	Red B – Médicos Suc1
Laptop1-WiFi-Suc1	DHCP (172.16.96.10+)	255.255.224.0	172.16.96.1	172.16.0.4	Red C – WiFi Suc1
Laptop2-WiFi-Suc1	DHCP (172.16.96.11+)	255.255.224.0	172.16.96.1	172.16.0.4	Red C – WiFi Suc1
PC1-RRHH-Suc2	172.16.160.2	255.255.224.0	172.16.160.1	172.16.128.4	Red A – RRHH Suc2
PC2-RRHH-Suc2	172.16.160.3	255.255.224.0	172.16.160.1	172.16.128.4	Red A – RRHH Suc2
PC3-Medico-Suc2	172.16.192.2	255.255.224.0	172.16.192.1	172.16.128.4	Red B – Médicos Suc2
PC4-Medico-Suc2	172.16.192.3	255.255.224.0	172.16.192.1	172.16.128.4	Red B – Médicos Suc2

Laptop1-WiFi	DHCP	255.255.224.0	172.16.224.1	172.16.128.4	Red C – WiFi
-Suc2	(172.16.224.10+)				Suc2
Laptop2-WiFi	DHCP	255.255.224.0	172.16.224.1	172.16.128.4	Red C – WiFi
-Suc2	(172.16.224.11+)				Suc2

Tabla 4 — Usuarios SMTP / FTP

Dispositivo	Usuario	Email	Contraseña	Usuario	Permisos FTP
	SMTP			FTP	
PC1-RRHH-Suc1	rrhh1	rrhh1@clinica.org	rrhh123	rrhh1	Lectura, Escritura, Eliminar, Listar
PC2-RRHH-Suc1	rrhh2	rrhh2@clinica.org	rrhh123	rrhh2	Lectura, Escritura, Eliminar, Listar
PC3-Medico-Suc1	medico1	medico1@clinica.org	med123	medico1	Solo Lectura, Listar
PC4-Medico-Suc1	medico2	medico2@clinica.org	med123	medico2	Solo Lectura, Listar
Laptop1-WiFi-Suc1	wifi1	wifi1@clinica.org	wifi123	wifi1	Lectura, Escritura, Listar
Laptop2-WiFi-Suc1	wifi2	wifi2@clinica.org	wifi123	wifi2	Lectura, Escritura, Listar
PC1-RRHH-Suc2	rrhh3	rrhh3@clinica.org	rrhh123	rrhh3	Lectura, Escritura, Eliminar, Listar

PC2-RRHH-Suc2	rrhh4	<u>rrhh4@clinica.org</u>	rrhh123	rrhh4	Lectura, Escritura, Eliminar, Listar
PC3-Medico-Suc2	medico3	<u>medico3@clinica.org</u>	med123	medico3	Solo Lectura, Listar
PC4-Medico-Suc2	medico4	<u>medico4@clinica.org</u>	med123	medico4	Solo Lectura, Listar
Laptop1-WiFi-Suc2	wifi3	<u>wifi3@clinica.org</u>	wifi123	wifi3	Lectura, Escritura, Listar
Laptop2-WiFi-Suc2	wifi4	<u>wifi4@clinica.org</u>	wifi123	wifi4	Lectura, Escritura, Listar

Tabla 5 — Registros DNS configurados

Nombre DNS	Tipo	IP Apuntada (Suc1)	IP Apuntada (Suc2)
web-clinica.org	A Record	172.16.0.5	172.16.128.5
medicos.clinica.org	A Record	172.16.0.6	172.16.128.6
rrhh.clinica.org	A Record	172.16.0.8	172.16.128.8
smtp.clinica.org	A Record	172.16.0.2	172.16.128.2
ftp.clinica.org	A Record	172.16.0.3	172.16.128.3
dns.clinica.org	A Record	172.16.0.4	172.16.128.4
dhcp.clinica.org	A Record	172.16.0.7	172.16.128.7

Tabla 6 — Configuración Access Points

Dispositivo	Conectado a	Puerto Switch	SSID	Autenticación	Contraseña
AP1-Suc1	Sw_1ro_Suc1	Fa0/21	Clinica-WiFi-Suc1	WPA2-PSK	clinica2025
AP2-Suc1	Sw_1ro_Suc1	Fa0/22	Clinica-WiFi-Suc1	WPA2-PSK	clinica2025
AP3-Suc1	Sw_1ro_Suc1	Fa0/23	Clinica-WiFi-Suc1	WPA2-PSK	clinica2025
AP4-Suc1	Sw_1ro_Suc1	Fa0/24	Clinica-WiFi-Suc1	WPA2-PSK	clinica2025
AP1-Suc2	Sw_1ro_Suc2	Fa0/21	Clinica-WiFi-Suc2	WPA2-PSK	clinica2025
AP2-Suc2	Sw_1ro_Suc2	Fa0/22	Clinica-WiFi-Suc2	WPA2-PSK	clinica2025
AP3-Suc2	Sw_1ro_Suc2	Fa0/23	Clinica-WiFi-Suc2	WPA2-PSK	clinica2025
AP4-Suc2	Sw_1ro_Suc2	Fa0/24	Clinica-WiFi-Suc2	WPA2-PSK	clinica2025

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC3-...	PC1-RRHH...	ICMP	Green	0.000	N	0	(e...	(delete)
	Successful	PC3-...	PC2-RRHH...	ICMP	Red	0.000	N	1	(e...	(delete)
	Successful	PC3-...	PC2-RRHH...	ICMP	Orange	0.000	N	2	(e...	(delete)

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.64.2

Pinging 172.16.64.2 with 32 bytes of data:

Reply from 172.16.64.2: bytes=32 time=10ms TTL=128
Reply from 172.16.64.2: bytes=32 time=1ms TTL=128
Reply from 172.16.64.2: bytes=32 time=1ms TTL=128
Reply from 172.16.64.2: bytes=32 time=1ms TTL=128

Ping statistics for 172.16.64.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 3ms

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```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping web-clinica.org

Pinging 172.16.128.5 with 32 bytes of data:

Request timed out.
Reply from 172.16.128.5: bytes=32 time<1ms TTL=127
Reply from 172.16.128.5: bytes=32 time=9ms TTL=127
Reply from 172.16.128.5: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.128.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 3ms

```

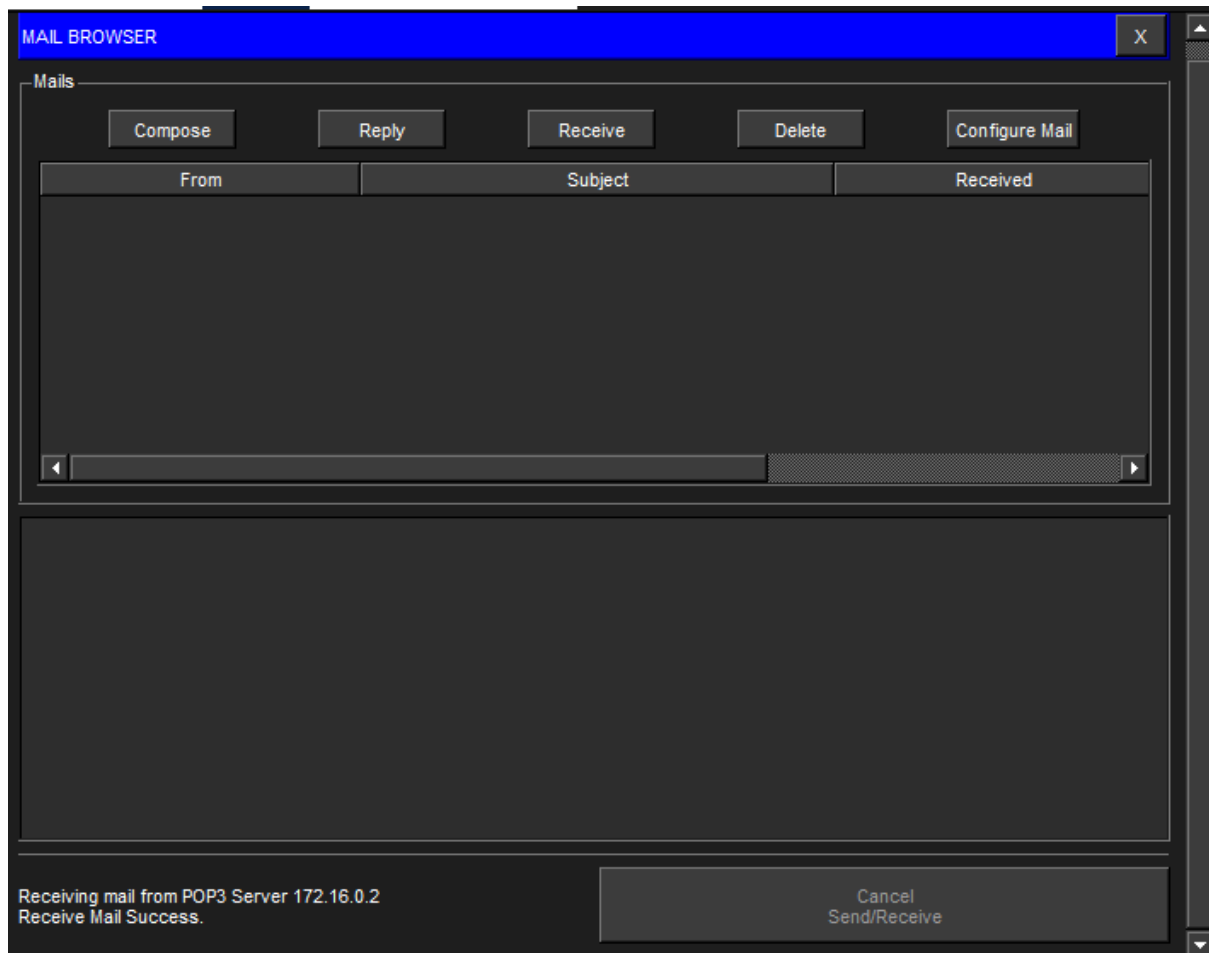


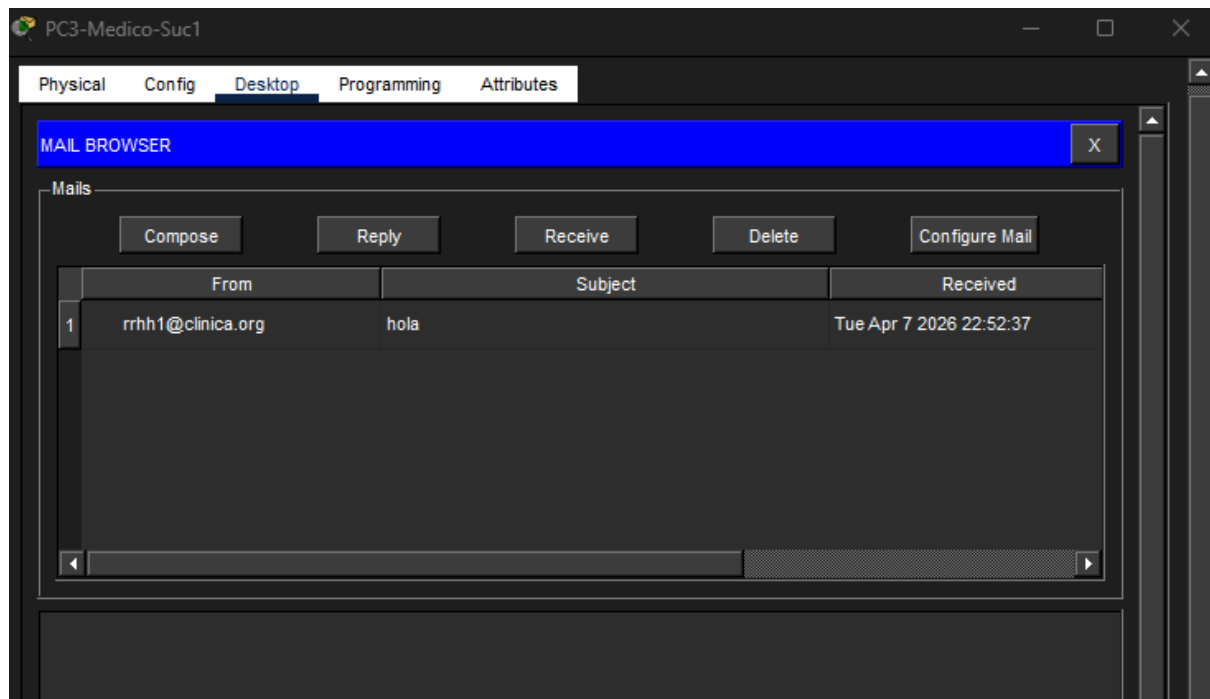
```
C:\>tracert 172.16.128.2

Tracing route to 172.16.128.2 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    172.16.64.1
  2  0 ms    0 ms    0 ms    10.255.0.2
  3  *        0 ms    0 ms    172.16.128.2

Trace complete.
```





```
C:\>ftp 172.16.0.3
Trying to connect...172.16.0.3
Connected to 172.16.0.3
220- Welcome to FT Ftp server
Username:rrhh1
331- Username ok, need password
Password:*****

%Error ftp://172.16.0.3/ (No such Account)
332- Need account for login
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```

C:\>ftp
C:\>ftp 172.16.0.3
Trying to connect...172.16.0.3
Connected to 172.16.0.3
220- Welcome to PT Ftp server
Username:rrhh1
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 172.16.0.3:
0   : asa842-k8.bin                               5571584
1   : asa923-k8.bin                               30468096
2   : c1841-advipservicesk9-mz.124-15.T1.bin      33591768
3   : c1841-ipbase-mz.123-14.T7.bin               13832032
4   : c1841-ipbasek9-mz.124-12.bin               16599160
5   : c1900-universalk9-mz.SPA.155-3.M4a.bin      33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin      33591768
7   : c2600-i-mz.122-28.bin                       5571584
8   : c2600-ipbasek9-mz.124-8.bin                 13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin     50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin      33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin             5571584
12  : c2800nm-ipbasek9-mz.124-8.bin               15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin      33591768
14  : c2950-i6q412-mz.121-22.EA4.bin             3058048
15  : c2950-i6q412-mz.121-22.EA8.bin             3117390
16  : c2960-lanbase-mz.122-25.FX.bin              4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin            4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin            4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin     8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin      10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin         33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin       83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin      505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG         159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG         184530138
26  : ie9k_iosxe.17.09.04.SPA.bin                596133776
27  : ir800-universalk9-bundle.SPA.156-3.M.bin     160968869
28  : ir800-universalk9-mz.SPA.155-3.M            61750062
29  : ir800-universalk9-mz.SPA.156-3.M            63753767
30  : ir800_yocto-1.7.2.tar                       2877440
31  : ir800_yocto-1.7.2_python-2.7.3.tar         6912000
32  : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
33  : pt1000-i-mz.122-28.bin                      5571584
34  : pt3000-i6q412-mz.121-22.EA4.bin            3117390

```

